

Hearing Silence

Neural state decoding, metacognitive awareness, and a research program for the inner experience that drives anxiety

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Hearing Silence — an independent research program

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Abstract

Background. Repetitive verbal thought — worry, rumination, self-critical commentary, and in some people voices experienced as not their own — is a transdiagnostic driver of anxiety and distress. The psychological mechanism that relieves it is well characterised: *decentering*, the capacity to experience a thought as a transient mental event rather than as a fact about the world. Treatment-related increases in decentering precede and predict symptom reduction across mindfulness-based treatment, CBT and acceptance-based therapies, with effect sizes for change in decentering itself of $d = 0.6$ – 1.85 . The skill is teachable but slow to acquire, and progress is currently measurable only by self-report.

Opportunity. Neural decoding has advanced sharply. Imagined sentences have been decoded in real time from intracortical arrays in four people at 26–54% word error rate over a 125,000-word vocabulary; non-invasive MEG reaches 32% character error rate; fMRI recovers the gist of imagined narratives for 41–74% of time points. Real-time fMRI of the posterior cingulate already separates “effortless awareness” from “distracted, efforting” awareness in meditators’ own reports, and scalp EEG detects mind-wandering at $AUC \approx 0.72$ within individuals.

Problem. The obvious application — transcribing the inner monologue — does not survive contact with the phenomenological evidence. In randomly sampled moments of everyday experience, inner speech occurs in 26% of samples and ranges from 0% to 75% between individuals; at rest, roughly 32% of sampled moments involve words at all. Content decoding is also modality-locked: a phoneme-based decoder is structurally unable to read signers, anendophasics, or people whose thought is predominantly imagistic or unsymbolised.

Approach. This paper argues that the tractable and ethically defensible target is the **state** that generates repetitive thought, not the words it produces. Inner speech is a motor simulation of articulation that emits a content-specific efference copy, and that prediction is what tags an utterance as self-generated. Within this framework, pathological worry is a **termination** failure with the self-tag intact, while a voice experienced as external is a **tagging** failure — two distinct faults in one mechanism, each separately measurable.

Aims. We propose the *Hearing Silence* program, with two deliverables: (1) a **Thought-Form Atlas**, the first corpus pairing high-density neural recording with validated moment-sampled phenomenology across verbal, visual, unsymbolised and kinaesthetic-signed experience; and (2) the **Attentional Return Index (ARI)**, a modality-independent, individually calibrated, real-time measure of how long a person dwells in self-referential processing and how quickly attention returns.

Significance. Eight sham-controlled randomised trials of attention neurofeedback have found no significant pooled effect, which means the field’s accumulated positive results cannot yet be attributed to neural specificity. A validated, sham-tested index of metacognitive state would convert contemplative training from something taken on faith into something that can be dosed, titrated and verified — and would give clinical research an objective endpoint for the inner experience that drives anxiety.

1. Introduction

1.1 The problem

The inner voice is not a neutral narrator. For a substantial fraction of people it is the medium through which anxiety is produced and sustained: anticipatory rehearsal that never resolves, self-criticism that repeats, and — in some — verbal content experienced as arriving from outside the self. The common thread is not the content of the words but the fact that the person cannot step back from them.

Contemporary treatment already targets this directly, and works. What it lacks is instrumentation. A clinician cannot observe whether a patient’s relationship to their own thought is changing except by asking them, and self-report is a lagging and unreliable indicator: in one controlled study, meditation training produced measurable network reorganisation **without** any corresponding increase in self-reported mindfulness. The subjective instrument moved after the brain did.

1.2 The reframe this paper argues for

The intuitive application of brain-computer interfaces here is transcription — read the monologue, show it to the person, let them see what they are saying to themselves. Sections 2, 7 and 8 set out why that target is the wrong one: inner speech is a minority of inner experience, varies between individuals from 0% to 75% of sampled moments, is absent entirely in some people, and runs in a non-verbal modality in others.

The target that survives scrutiny is the **state**, not the content. Specifically: whether a person is currently absorbed in self-referential verbal processing, how long they remain there, and how quickly attention returns. That state is modality-independent, it is the variable the psychological mechanism actually operates on, and it can be measured without ever reconstructing a word.

This reframing is not a retreat from ambition. It makes the technology applicable to everyone rather than to the subset whose inner life happens to run in phonemes, and it removes the single largest ethical hazard — a device that reports what someone is thinking.

It also has a concrete clinical form, which is worth stating plainly before the evidence sections. The intervention is a light EEG band and a haptic cue: when the decoder detects self-referential absorption, the wearer receives a tactile pulse within a second — no words, no transcript, no content of any kind. The participant surfaces and re-engages. Repeated across sessions, the external cue is expected to train the endogenous detector described in §3.3, which is why H6 tests whether benefit survives device removal.

1.3 Research goal

To make metacognitive awareness measurable.
Specifically: to establish an objective, individually calibrated, modality-independent index of self-referential absorption and attentional return, validated against moment-sampled experience, and to determine whether delivering that index back to a person in real time accelerates the acquisition of decentering beyond what practice alone achieves.

Two deliverables follow from that goal, and the program in Section 12 is organised around them.

Deliverable 1 — the Thought-Form Atlas. The first corpus pairing high-density neural recording with validated moment-sampled phenomenology, covering the full observed range of inner experience: inner speaking, inner seeing, unsymbolised thinking, feeling, sensory awareness, and — in deaf signing participants — kinaesthetic inner signing. No such corpus exists. Every published inner-speech dataset uses **cued** trials, in which a word is presented and the participant imagines it, which tells us nothing about spontaneous thought.

Deliverable 2 — the Attentional Return Index (ARI). A real-time index with three components: *dwell* (how long a self-referential episode lasts), *return latency* (how long after onset attention re-engages), and *probe-caught fraction* (how often the person notices before any external cue). The ARI is designed to be computable from scalp EEG, to be calibrated per individual, and to be reported without any content reconstruction.

1.4 Why this is a wellness problem and not only a clinical one

The populations in Section 6 differ sharply in mechanism and risk, and the strongest case is the least medicalised one. Anxious inner speech is dimensional — it is present in diagnosed anxiety and depression and in people with no diagnosis at all. An index of attentional return is therefore not a diagnostic instrument but a training signal, closer in spirit to heart-rate variability than to a biomarker.

That framing sets the standard this program holds itself to: the output is something a person uses on themselves, never something inferred about them by a third party.

2. The state of inner-speech decoding

Inner speech is decodable from implanted electrodes, imperfectly, and from MEG, worse. Every result below comes from a **cooperating** participant producing speech **on cue**. No published work decodes spontaneous, uninstructed thought at usable rates.

STUDY	SIGNAL	PARTICIPANTS	WHAT WAS DECODED	VOCABULARY	BEST RESULT
Willett et al. 2023, <i>Nature</i>	4 intracortical arrays, areas 6v + 44	1 (ALS)	Attempted speech	50 words / 125,000	9.1% WER / 23.8% WER, 62 wpm
Metzger et al. 2023, <i>Nature</i>	High-density ECoG, sensorimotor cortex	1 (brainstem stroke)	Attempted speech → text, voice, avatar	1,024 words (sim. to 39,378)	78 wpm, 25% WER median
Wandelt et al. 2024, <i>Nat Hum Behav</i>	Intracortical, supramarginal gyrus + S1	2 (tetraplegia)	Internal speech	8 words	79% online (P1); 23% (P2)
Kunz et al. 2025, <i>Cell</i>	Intracortical, motor cortex	4 (ALS / stroke)	Imagined sentences	50 words / 125,000	14–33% WER / 26–54% WER
Wairagkar et al. 2025, <i>Nature</i>	256 microelectrodes, ventral precentral gyrus	1 (ALS)	Attempted speech → voice	Open	10 ms latency; transferred to <i>mimed</i> speech untrained
Tang & Huth 2023, <i>Nat Neurosci</i>	fMRI, ~16 h training per subject	3 (healthy)	Imagined stories (semantic gist)	Open	41–74% of time points
Brain2Qwerty 2025	MEG / EEG, non-invasive	20 MEG, 20 EEG	Typed sentences	Open	32% CER (MEG); 67% (EEG)
Nieto et al. 2022, <i>Sci Data</i>	128-ch EEG	10 (healthy)	Inner speech, 4 direction words	4 words	Dataset only; no classifier reported

2.1 Four findings that constrain any application

Inner speech occupies motor cortex as an attenuated copy of attempted speech. Kunz et al. report that inner and attempted speech activate overlapping populations with similar patterns, but that inner speech shows *weaker magnitude*. Wandelt et al. found 71–79% of neurons tuned during internal speech were also tuned during vocalised speech — “similar but distinct.” The same decoder architecture therefore transfers, and transfers poorly.

Uninstructed inner speech is already detectable. Kunz et al.’s decoder captured speech participants were never instructed to produce: numbers generated while silently counting shapes on a screen, and rehearsal sequences during a memory task. Spontaneous inner speech is visible to a decoder trained only on cued speech — which is simultaneously the strongest evidence that the approach could generalise, and the clearest statement of its privacy hazard.

Decoding collapses under non-cooperation. Tang & Huth tested adversarial strategies directly: counting by sevens reduced recovery to 4–50%, naming animals to 0–3%, and imagining a different narrative to 1–26%. Cross-subject decoders recovered 1–5% of time points against 72–82% within-subject. Cooperation is required both to train and to run a decoder.

Non-invasive content decoding is two orders of magnitude behind, and MEG is not portable. A 32% character error rate from MEG is a laboratory result obtained inside a magnetically shielded room with a cryogenic sensor array. EEG, the portable modality, reaches 67% — unusable for content. For **state** decoding, however, EEG is adequate, which is the central engineering argument of this paper.

2.2 What has not been attempted

Implanted-array research remains confined to clinical-trial participants with paralysis or ALS, because the risk-benefit calculation closes only when the participant gains restored function. Intracranial recording in cognitively intact people is nonetheless routine: patients with drug-resistant epilepsy are implanted with depth electrodes for one to two weeks of clinical monitoring, and research protocols are commonly added to their consent. This is the access route for any intracranial phase of the present program, and it is addressed in Sections 11 and 12.

3. The psychological target: decentering, not silence

3.1 What observing a thought actually does

The therapeutic mechanism is not thought removal. It is **decentering**: a change in how a thought is held, not whether it occurs.

The operational definition is specific — “a change in emotional reactivity towards negative inner events... a shift in one’s awareness away from the thematic content of those events, towards the cognitive activities underlying these events” (*Translational Psychiatry*, 2021). The canonical example contrasts registering “*I am having the thought that I am depressed*” with “*I am depressed*.” The thought still arrives; its grip changes.

Treatment-related increases in decentering **precede and predict** symptom reduction across mindfulness-based treatment, CBT and acceptance-based therapies, with effect sizes for change in decentering itself of $d = 0.6$ – 1.85 . Decentering — not relaxation, and not thought suppression — is the active ingredient any closed-loop system should be trying to amplify.

3.2 Why “silence” is the wrong objective

The colloquial goal of a quiet mind is better specified as **low-cost attentional return**: thoughts still arise, they are noticed sooner, attention re-engages faster, and the noticing stops requiring effort.

This distinction determines the success metric. A silence objective predicts that default-mode activity should fall and stay low. A return objective predicts something different and more tractable: that *dwell time* in self-referential states should shorten while their *frequency* may barely change.

The evidence favours the second. Vago & Zeidan (2016) argue explicitly **against** treating mindfulness and mind-wandering as opposite poles of a single axis, proposing instead “skillful toggling back and forth between conceptual and nonconceptual processes.” A system that rewards default-mode suppression would also be suppressing autobiographical memory, prospection and social reasoning, which share that network.

3.3 The control architecture training actually produces

The most directly relevant finding for system design is Bremer et al. (2022). In 46 meditation-naïve adults, one month of training did not flatten the default mode network — it *increased* connectivity between the default mode, salience and central executive networks.

The authors’ interpretation is a control-systems description. The salience network monitors; it detects the slip into self-referential processing; it signals the executive network; attention is restored. Training makes people “more effectively aware of DMN-related processes” and able to “regain attention more efficiently.”

That is a detector feeding an actuator, and it identifies precisely where an external system can contribute. **A closed-loop decoder would act as a prosthesis for the detector stage** — an outboard salience signal firing on the lapse before the endogenous monitor catches it. Critically, those connectivity changes appeared *without* increases in self-reported mindfulness, which establishes neural signal as a leading indicator and self-report as a lagging one.

3.4 The signal is already readable in real time

Garrison et al. (2013) delivered real-time fMRI feedback from the posterior cingulate cortex to experienced meditators and matched it against their moment-to-moment reports. PCC activity fell during undistracted awareness, concentration, observing sensory experience, contentment and “not efforting.” It rose during distraction, interpreting, efforting and discontentment.

A mapping between one measurable region and exactly the subjective distinction this program targets therefore already exists, in real time, in individual people. Its limitations are equally clear: experienced meditators, a 3-tesla scanner, and a haemodynamic signal that is slow and regionally non-specific. Deriving a PCC-equivalent index from portable hardware is the first engineering problem the program must solve.

4. Closed-loop metacognitive training

Every component of the loop has been built separately; none has been assembled with a modern decoder at the detector stage. Setting out the stages shows how little of the system is novel — and how little of it a device can perform on the user’s behalf.

Only the detector stage is new technology

Closed-loop metacognitive training, one cycle

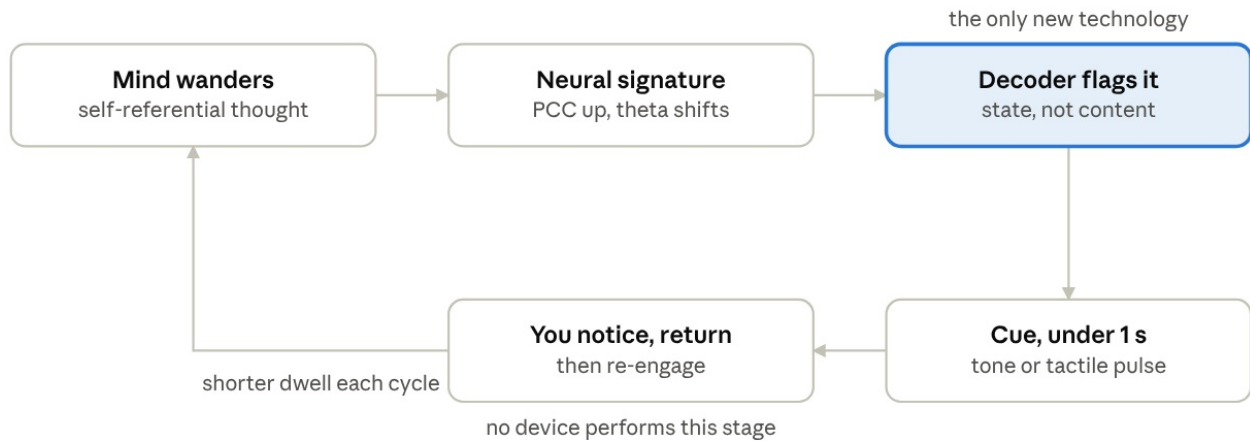


Figure 1. Closed-loop metacognitive training: five stages, one of which is new technology.

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Stages 1, 2 and 5 occur in every person already. Stage 3 is the engineering problem; stage 4 is a design problem concerning cue modality and latency.

4.1 Systems that exist

Chen & Ziegler (2025) review closed-loop systems built for meditation and mindfulness. fMRI implementations target posterior cingulate deactivation, default-mode / central-executive / salience interaction, and a “positive diametric activity” index computed as central-executive minus default-mode activation. EEG implementations target alpha (8–13 Hz) and theta (4–8 Hz) as indices of relaxed alertness, frontal midline theta for executive control, and gamma suppression over default-mode regions. Consumer headbands have demonstrated feasibility with improved attentional performance.

Their own system, MediTrain, adapts trial difficulty to real-time performance and produced sustained-attention gains and connectivity changes in adolescents with early-life adversity. It closes the loop on *behaviour*, not on neural signal.

4.2 The result the field has not yet answered

A systematic review and meta-analysis of neurofeedback for attention in healthy adults pooled 15 studies and 569 participants (*Imaging Neuroscience*, 2024). Against no intervention, the effect was small but significant: $SMD = 0.27$ (95% CI 0.10–0.44).

Against **sham** neurofeedback, across eight randomised controlled trials, there was **no significant pooled effect**.

This is the central unresolved problem in the field, and it defines the evidentiary bar for the present program. It implies that most measured neurofeedback benefit to date is attributable to practice, attention, expectancy and experimenter contact rather than to the specificity of the neural signal. Accordingly, every interventional study proposed in Section 12 carries a **yoked sham arm** from first design: feedback derived from another participant’s recording, or from a time-shifted segment of the participant’s own, matched for difficulty and reinforcement rate.

4.3 What a state decoder adds over existing systems

Three specific contributions, none of which requires content reconstruction.

1. **Specificity of the signal.** Existing systems report an arousal or band-power level. A decoder trained on moment-sampled phenomenology can distinguish *verbal self-talk* from *imagery* from *bodily sensation* — distinctions the sampling literature shows vary substantially between individuals. Feedback that identifies the mode of experience is a categorically different intervention from feedback that reports an amplitude.
2. **Labels without interruption.** Experience sampling requires stopping the participant to ask. A decoder validated against sampling can label the unsampled moments in between, which is the only route to a corpus large enough to train on.
3. **Latency.** Spontaneous metacognitive insight arrives seconds to minutes after a lapse. A detector firing in under a second converts the learning problem from delayed to near-immediate reinforcement. That, rather than content transcription, is the plausible mechanism by which instrumentation could accelerate what contemplative practice achieves slowly.

5. Existing data resources

The first phase of this program requires no hardware, no recruitment and no device exemption. Six public datasets span the argument, and two of them are close to purpose-built for the research goal in Section 1.3.

DATASET	MODALITY	SCALE	WHAT IT GETS YOU	ACCESS
EEG meditation study (Delorme & Brandmeyer)	64-ch EEG	24 participants, 3 sessions, 5.69 GB	Start here. Meditation with probes every ~2 min rating concentration and mind-wandering — labelled lapses in the population you care about	OpenNeuro ds001787 / NEMAR, CCO
EEG–fMRI with online experience sampling	31-ch EEG + fMRI	24 adults 18–35, ~47 sessions, 468 thought ratings each	Then here. 13 dimensions of thought content rated at rest, <i>plus</i> STAI-S, GAD-7 and PHQ-9 — so you can test whether thought-type decoding tracks anxiety	OpenNeuro ds007216
Thinking Out Loud (Nieto et al. 2022)	128-ch EEG, 1024 Hz	10 participants, ~5,640 trials	The standard inner-speech EEG benchmark: 4 Spanish direction words across inner, pronounced and visualized conditions	OpenNeuro ds003626, BIDS
Chisco (2024)	125-ch EEG, 1000 Hz	3 participants, 6,681 sentences, 20,043 trials, 900+ min each	Sentence-scale imagined speech across 39 semantic categories — the only EEG set with this depth per subject	OpenNeuro ds005170, CCO
Bimodal EEG–fMRI inner speech (2023)	64-ch BioSemi + fMRI	4 usable participants, 320 trials per modality	Cross-modal check: what EEG sees of what fMRI sees, on 8 words in two semantic categories	OpenNeuro, BIDS
Brain Treebank (Wang et al. 2024)	Intracranial sEEG	10 patients, 1,688 electrodes, 43.5 h, 223,000 words	The invasive signal, free: naturalistic film-watching with word-level timing and dependency parses	BrainTreebank.dev , CC BY 4.0

5.1 Interpretation

The first two rows bear directly on the research goal, and neither is currently used in the BCI literature. The inner-speech datasets in rows 3–5 are all **cued**: a word is presented, the participant imagines it, the trial ends. They are informative about decodability and silent about spontaneous thought.

Brain Treebank establishes that intracranial data from cognitively intact people is publicly available — 1,688 electrodes across 10 participants, released under CC BY 4.0 — which removes any need for new invasive recording during the early phases of this program.

5.2 The dataset that does not exist

The resource the research goal requires is **simultaneous high-density neural recording and Descriptive Experience Sampling with trained participants, at rest, across many sessions**. [Hurlburt et al. \(2015\)](#) came closest — 5 participants, 180 in-scanner samples, nine sessions of 25 minutes each — and that study was qualitative in design rather than a decoding corpus.

Constructing that corpus is the Thought-Form Atlas (Deliverable 1), and it requires no implant. Descriptive Experience Sampling is laborious: participants must be trained iteratively across days, and the same study found retrospective questionnaires diverging from moment-sampled experience by up to 81%. That labour is precisely why the corpus does not yet exist, and why building it constitutes a contribution independent of whether the closed-loop intervention ultimately succeeds.

6. Three clinical targets, distinguished

The colloquial descriptions — “voices in my head,” “a mind that will not stop” — cover at least three conditions with different mechanisms, different evidence bases and different risk profiles. Treating them as one phenomenon is both a methodological error and a safety hazard.

	ANXIOUS INNER SPEECH	AUDITORY VERBAL HALLUCINATIONS	DISSOCIATIVE PLURALITY
What it is	Rumination, worry loops, harsh self-talk; experienced as <i>mine</i>	Voices experienced as <i>not mine</i> , often with auditory quality and location	Experienced sense of distinct selves with their own perspectives and memory claims
Where it breaks	Termination. The loop runs past usefulness; the self-tag is intact	Self-tagging. The efference copy is degraded or mistimed, so the utterance arrives untagged	Not characterized — no validated mechanistic model exists
Where it sits	Dimensional; present in anxiety and depression and in healthy people	Psychotic disorders, but also reported by people with no psychiatric illness	Dissociative disorders; also a non-clinical plural community that does not seek treatment
Mechanism evidence	Strong. Decentering mediates improvement, $d = 0.6\text{--}1.85$	Mixed. Meta-analysis finds no significant Broca’s or Heschl’s activity during AVH; left insula is the consistent finding	Weak. 9 functional imaging studies in 40 years; the critical review finds they “do not support firm claims”
What a decoder could offer	State detection and faster attentional return — the strongest case	Possibly content, for the narrow case where the voice’s words drive distress or risk	Essentially nothing yet; the measurement model isn’t settled
Main risk	Over-pathologizing normal thought	Feeding distressing content back to someone with impaired reality monitoring	A device presented as adjudicating whether someone’s selves are “real”

6.1 Evidence on auditory verbal hallucinations

The inner-speech model of AVH — that voices are the speaker’s own inner speech misattributed to an external source — is the natural bridge between decoding and clinical application. It is only partly supported. A [review in *Translational Psychiatry*](#) reports that meta-analyses found **no significant activity at Broca’s area or Heschl’s gyrus during hallucinations**, and no structural abnormality in the arcuate fasciculus. What survives is the left insula: active during AVH and structurally reduced in hallucinators, consistent with a misattribution account rather than an overproduction account.

The practical consequence is direct. If voices are not generated by the inner-speech production system, a decoder trained on motor-cortex inner speech will not read them. Section 9 proposes the mechanism that reconciles this: AVH as a failure of self-tagging rather than of production.

Real-time fMRI neurofeedback for AVH has been attempted in six proof-of-concept studies targeting superior temporal gyrus, anterior cingulate, paracingulate cortex, default-mode nodes and inferior frontal gyrus, with reported symptom reductions. The review's own appraisal notes small underpowered samples, no randomised controlled trials, unclear durability and movement artefact occurring *during* hallucinations.

6.2 Evidence on dissociative plurality

Schlesselmann et al. (2026, *Memory*) reviewed every functional neuroimaging study of dissociative identity disorder conducted over 40 years and identified nine. Their conclusion is that functional imaging across identity states “does not support firm claims about identity fragmentation or inter-identity amnesia, nor does it decide between trauma-based and sociocognitive accounts.” The limitations are structural: absent clinical comparison groups, unaccounted comorbidity, and reverse inference.

This does not bear on whether the experience is real to those reporting it. It establishes that no validated neural target exists, and therefore that a decoder has nothing to aim at. A device presented as adjudicating the question would exceed what the instrument can settle.

6.3 Target selection

This program begins with **anxious inner speech in people without a psychiatric diagnosis**. That population has the strongest mechanism evidence, the lowest risk profile, the largest accessible sample and the most straightforward regulatory path. A closed-loop trainer that cannot outperform sham in healthy anxious adults will not help people with psychotic or dissociative disorders, and the harder populations should be approached only after the easier case is demonstrated.

7. Inner experience that is not speech

7.1 How much of thought is verbal

Randomly sampled moments of everyday experience are predominantly non-verbal. Across 295 samples from 30 participants, Heavey & Hurlburt (2008) recorded inner seeing in 34% of moments, inner speech in 26%, feelings in 26%, sensory awareness in 22% and **unsymbolised thinking** — definite thought content with neither words nor images — in 22%. Inner speech ranged from **0% to 75% between individuals**.

Sampled at rest inside a scanner, the pattern holds: sensory awareness averaged 59% of moments, inner speaking 29%, unsymbolised thinking 20%, and only about 32% of moments involved words at all (Hurlburt et al., 2015).

Two consequences follow. A transcription-oriented decoder aimed at a resting mind is idle for most of its operating time. And unsymbolised thinking offers it no target at all: there are no phonemes to classify and no image to reconstruct. Since every current open-vocabulary decoder pairs a weak neural signal with a strong language model, such a system pointed at wordless thought will generate fluent, confident, fabricated sentences and attribute them to the person.

7.2 Populations a speech decoder structurally excludes

- **Anendophasia** — absence of the experience of inner speech. Nedergaard & Lupyan (2024) compared 46 low-inner-speech with 47 high-inner-speech adults: the low group performed worse on verbal working memory and rhyme judgment, while task-switching and categorical perception were unaffected. The absence is measurable and consequential, not a reporting artefact.
- **Anauralia** — absence of auditory imagery. In a New Zealand survey of over 32,000 people rating auditory imagery vividness from 1 to 7, roughly 1 in 100 scored zero, while about 25% were “hyperauralic.” It frequently co-occurs with aphantasia (University of Auckland, 2025).
- **Aphantasia** — absence of visual imagery. An image-based decoder fails this group for the same structural reason a speech decoder fails anendophasics.

Screening by neuroimaging does not resolve the problem. In the Auckland work, MRI and EEG showed **similar activation patterns in silent and non-silent minds** during sound-related tasks, so activation alone does not identify which group a participant belongs to.

7.3 Deaf signers: inner language in a different modality

The widespread assumption that deaf people think in images is not supported by the available evidence, and the better-supported account has direct engineering implications.

Gregory (*Synthese*, 2026) argues that deaf signers do have inner speech, but that it differs fundamentally from auditory inner speech. Three candidate forms are considered: visual imagery of signs, **kinaesthetic imagery** (the felt sensation of signing), or a hybrid that is predominantly kinaesthetic with secondary visual content. The analysis favours the kinaesthetic, partly on the grounds that visual inner signing would be *perspectival* in a way auditory inner speech is not — signers see their own hands from behind and an interlocutor's from the front, so a visual inner sign must adopt a viewpoint.

The empirical base is thin: electromyographic work from 1937, working-memory errors reflecting visual-gestural rather than phonological properties (1975), and a 1997 PET study showing mental signing activates language regions. Gregory's own assessment is that there are **no empirical data** on signed inner-speech phenomenology and no systematic data on the distribution of forms across signers.

7.4 A testable architectural consequence

If inner signing is kinaesthetic and manual, the relevant motor representation is **hand and arm, not orofacial** — and the highest-performing text BCI yet built reads exactly that region. Willett et al. (2021) implanted two arrays, 192 electrodes, in the hand “knob” of precentral gyrus in a single tetraplegic participant imagining handwriting, achieving **90 characters per minute at 94.1% raw accuracy and above 99% with a language model**.

The mechanism underlying that result predicts the extension. Handwritten letters carry **2.65 times the temporal dimensionality** of straight-line point-to-point movements, and that temporal complexity renders neural trajectories *more* separable rather than less. Signs are at least as temporally complex as handwritten characters.

The resulting prediction is specific and untested: **imagined or attempted signing should decode from hand-area arrays at least as well as imagined handwriting, and better than imagined speech decodes from orofacial cortex**. This appears as H9 in Section 13. For a deaf BCI user, such a system would constitute a native-language interface rather than a translation into another population's modality.

7.5 Design principle

Content decoding is **modality-locked**. A phoneme decoder reads phoneme-users; a sign decoder reads signers; an image decoder reads visualisers. Each serves the subset of people whose inner life runs in that format and returns noise, or fabrication, for everyone else.

State decoding is **modality-independent**. Absorption in self-referential processing is a state that a signer, a visualiser and an anendophasic can each occupy, and it can be measured without determining which. For a goal defined in terms of population-wide wellness, this is not a fallback position; it is the only layer that generalises.

8. Constraints on what can be measured

Five constraints bound any system of this kind. The first two are not engineering problems and will not be resolved by improved electrodes; they shape the study design proposed in Section 12.

8.1 No ground truth at rest. Every decoding result cited in Section 2 derives from cued trials with a known label. Spontaneous thought carries no label. The only rigorous labelling method, Descriptive Experience Sampling, requires interrupting the participant, and requires training them across multiple days to report accurately — which is itself closely related to the intervention under test. Measurement instrument and treatment are partially confounded by construction, and the design must separate them explicitly through an active control that receives sampling training without feedback.

8.2 The decoder requires cooperation precisely when cooperation is least available. Decoding collapses under mental resistance, which is a genuine privacy protection. It is also an obstacle: the target state is absorbed, unaware rumination, during which the participant is not deliberately holding still for a decoder. Any system requiring active cooperation misses its target state by definition. This is a further argument for state over content decoding, since state indices do not depend on compliance with a production task.

8.3 Nothing generalises across individuals. Cross-subject decoders recovered 1–5% of time points against 72–82% within-subject in fMRI semantic decoding. In intracortical internal-speech decoding, the same paradigm and implant produced 79% accuracy in one participant and 23% in the other. Every model should be assumed to require hours of that individual's own labelled data, which places a hard floor under the cost of the intervention and makes per-person calibration a design requirement rather than a refinement.

8.4 Chronic hardware performs better than once feared, but not reliably. Across 14 BrainGate participants and 20 Utah arrays followed for up to 7.6 years, mean electrode yield was 35.6%, declining only 7% from the first three months to the last, and 11 of 14 arrays supported usable decoding throughout ([2025 preprint](#)). However, 3 of 14 never produced a consistent decoding signal. An approximately one-in-five probability that a technically successful implantation yields an unusable array is a material consideration for any invasive phase.

8.5 Silence is the wrong objective function. The default mode network also supports autobiographical memory, prospection and social reasoning. A reward signal defined as reduced default-mode activity trains suppression of faculties the program has no interest in suppressing. Vago & Zeidan argue for toggling rather than suppression, and Bremer et al. found training *increased* inter-network connectivity. An objective function defined as silence optimises toward blunting rather than clarity, which is why the Attentional Return Index is defined over dwell and return latency rather than over amplitude.

9. How inner speech is generated

The question of which neural system produces self-directed verbal thought, and how, has a more specific answer than “the left inferior frontal gyrus.” That answer is the theoretical core of this program, because it specifies two separate failure points and therefore two separate measurement targets.

9.1 Inner speech is a motor act, not a remembered sound

[Nalborczyk et al. \(Cortex, 2023\)](#) applied TMS over the lip region of primary motor cortex in 31 participants and recorded motor evoked potentials in orofacial muscles while they covertly produced either /bu/ (lip-intensive) or /gi/ (minimal lip involvement).

Inner **speaking** raised motor cortex excitability more than inner **hearing**, and most strongly for the lip-intensive syllable. Inner speaking is a motor simulation of articulation; inner hearing is retrieval of stored sensory content with minimal motor engagement. Ordinary language treats both as “the voice in my head”; the brain implements them differently.

This explains why intracortical speech decoders locate inner speech in motor cortex at all, and why it presents there as an attenuated copy of attempted speech rather than as a separate signal.

9.2 It carries a content-specific prediction of its own sound

[Whitford et al. \(eLife, 2017\)](#) had 42 participants imagine the phoneme /ba/ or /bi/, time-locked by an animated cue to an audible probe /BA/ or /BI/. The auditory N1 was significantly smaller when the imagined phoneme **matched** the heard phoneme than when it mismatched: $t(41) = 2.54, p = .015$.

Sensory attenuation depended on content match, not merely on the act of imagining. Inner speech therefore emits a time-locked, **content-specific efference copy** — a forward model predicting what that particular inner utterance would sound like.

A subsequent study with 70 participants excluded the principal alternative explanation: priming a syllable visually *without* producing inner speech yielded no N1 suppression, no P2 change and no gamma change ([eNeuro, 2026](#)). Production is necessary; the effect reflects corollary discharge rather than expectancy.

9.3 Two generative mechanisms, not one

An [activation likelihood estimation meta-analysis in *NeuroImage* \(2023\)](#) concludes that the classical account — inner speech as truncated overt speech implemented in left inferior frontal gyrus — is insufficient alone. The framework consistent with the imaging data is dual:

- a **corollary discharge** mechanism in speech *production* regions, supporting deliberate, self-authored inner speech
- a **perceptual simulation** mechanism in speech *perception* regions, supporting inner speech with a more sensory, other-voiced quality

Distinct phenomenological varieties engage distinct pathways. Which pathway is active is therefore itself a measurable property of an episode, and a candidate component of the Attentional Return Index.

Worry and voices are two failures of the same self-tagging step

How inner speech is generated, and the two places it goes wrong

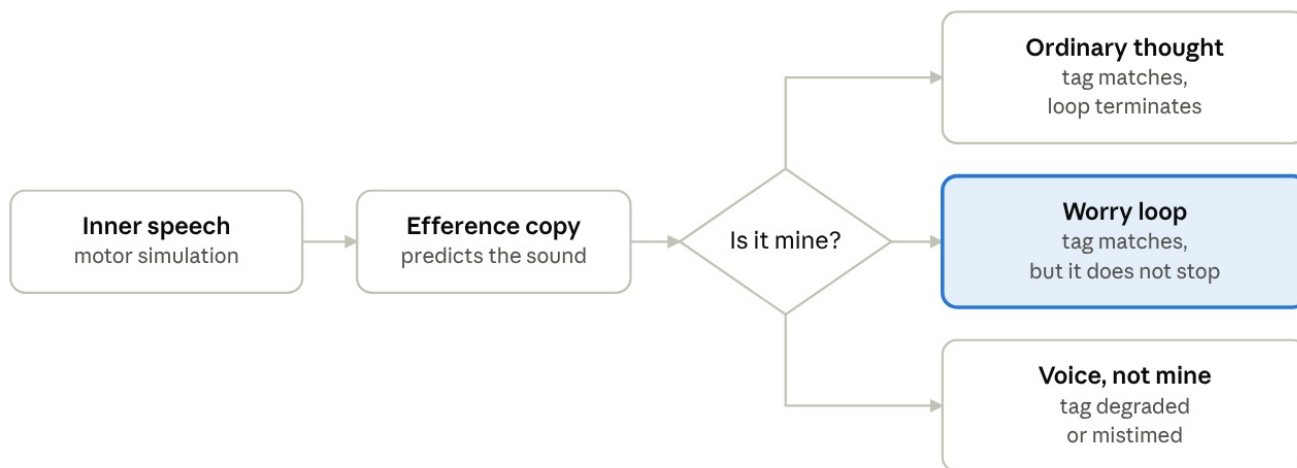


Figure 2. Worry and voices as two failures of the same self-tagging step.

Figure 2. Worry and voices as two failures of the same self-tagging step.

The branch point is identical in all three cases. Pathological worry is a **termination** failure with the self-tag intact; a voice experienced as external is a **tagging** failure. This accounts for why they are phenomenologically distinct, why they respond to different interventions, and why the three populations in Section 6 should not be pooled.

9.4 Why the efference copy is the program’s most useful construct

The efference copy functions as a **self-tag**. Inner speech normally arrives pre-labelled as the person’s own because their own forward model predicted it. Three consequences follow:

1. **It separates voices from thoughts mechanistically.** If the efference copy is degraded or mistimed, the utterance arrives untagged and the only available attribution is external agency. This fits the AVH evidence better than the simple inner-speech model, including the left insula finding and the absence of Broca’s and Heschl’s activation during hallucinations.
2. **It yields an index requiring no content decoding.** N1 suppression is an event-related potential effect obtainable with a 32-channel EEG system and an appropriate paradigm. Nothing about the content of the utterance need be recovered.
3. **It provides a within-person marker of deliberateness** — the dimension that distinguishes ordinary anticipatory rehearsal from intrusive rumination, and the dimension the ARI must capture.

9.5 Terminological precision this framework requires

Worry is not “voices” in the clinical sense. Worry is verbal inner speech, self-authored and experienced as the person’s own. Auditory verbal hallucinations are experienced as *not* the person’s own, and meta-analysis finds no significant Broca’s or Heschl’s activity during them. The two share machinery — both run through the inner-speech and efference-copy system — but constitute different failure states of it. The precise formulation, and the one this program adopts, is that **worry is inner speech whose self-tag is intact but whose termination signal has failed.**

Elimination is not a viable therapeutic target. [Wang, Hagger & Chatzisarantis \(2020\)](#) meta-analysed 31 studies employing the thought-suppression paradigm. **Rebound effects were observed regardless of cognitive load** — suppressed thoughts return with greater accessibility once suppression ceases — while immediate enhancement during suppression appeared only under cognitive load.

This compounds with the cognitive-avoidance account of worry ([Borkovec, Ray & Stöber, 1998](#)), under which worry is *already* avoidance: verbal thought deployed to damp imagery and somatic fear, negatively reinforced by the immediate relief it produces while blocking the emotional processing that would extinguish the fear. Suppressing worry adds avoidance to avoidance, with rebound attached. Every evidence-based treatment in this area moves toward the aversive material rather than away from it.

The program’s targets are therefore the three modifiable properties of the loop: **onset** (what initiates it), **maintenance** (what sustains it past utility), and **termination** (what permits it to stop). All three are measurable from a state decoder, and none requires that the inner voice be removed.

10. From correlation to mechanism

10.1 What localisation can and cannot establish

The question most often asked of this technology — which brain region is responsible for anxious self-talk — does not have the kind of answer it appears to request, and the literature is explicit about why.

A [systematic review of 24 studies](#) of repetitive negative thinking and brain morphology (Demnitz-King, Göehre & Marchant, 2021) found associations in dorsolateral prefrontal cortex, anterior cingulate cortex and the superior longitudinal fasciculus, then concluded that “inconsistent associations... preclude specific conclusions being reached regarding any one region’s association with RNT.” The authors further note they could not determine whether the structural differences *precede* repetitive negative thinking or *result from* it.

Three questions are routinely conflated and should be separated:

1. *What is the person thinking?* — decodable in principle, poorly, and only under cooperation.
2. *What neural activity accompanies it?* — measurable, correlational, and regionally inconsistent across studies.
3. *Why does this person hold this belief?* — a question about learning history, reinforcement and development. No imaging method answers it, and a system that claimed to would be fabricating.

Causal claims require perturbation or longitudinal design: stimulation during intracranial monitoring, TMS, or prospective cohorts. Demnitz-King et al.’s own recommendation is high-quality longitudinal work to establish direction of effect.

10.2 The mechanistic hypothesis worth testing

The cognitive-avoidance account of worry supplies a mechanism that is both specific and testable within a single individual. [Borkovec, Ray & Stöber \(1998\)](#) propose:

1. Worry is **predominantly verbal-linguistic** rather than imaginal. Mental-content sampling during worry induction shows a marked predominance of thought over images, in both patients with generalised anxiety disorder and controls.
2. Verbal thought is less tightly coupled to affective and physiological response systems than imagery. Worrying therefore **suppresses** cardiovascular response to threatening images.
3. That suppression constitutes immediate relief, negatively reinforcing the worry while blocking the emotional processing that would extinguish the underlying fear.
4. Successful therapy normalises the thought-to-image ratio.

If this account is correct, the verbal channel is not a symptom of anxiety but its **mechanism** — which makes an inner-speech-sensitive state decoder precisely targeted for anxiety, and poorly targeted for hallucination and dissociation. This inverts the intuitive ordering of the three populations in Section 6 and provides the strongest theoretical justification for the target selection in §6.3.

10.3 A tractable test of temporal precedence

The causal question can be made answerable by reframing it as a within-person ordering question: **does the verbal-rehearsal state precede physiological arousal, or follow it?**

The proposed design records EEG together with autonomic measures — heart-rate variability and skin conductance — during anticipated social evaluation, and tests Borkovec’s specific prediction: that the onset of verbal rehearsal is followed by somatic *suppression* rather than somatic increase. This appears as H4 in Section 13.

The study requires no implant and no device exemption, only a standard behavioural protocol. A reliable finding that verbal rehearsal precedes and then damps a somatic response within individuals would identify a within-person mechanism — which is as close to a causal account as current methods permit, and which has not been attempted with modern state-decoding methods.

11. Ethics, governance and participant protection

11.1 Neural data is now a regulated category

The regulatory position changed materially between 2024 and 2026. [Colorado](#) legislated first in 2024, adding neural data to “sensitive data” under its consumer privacy law and requiring **opt-in** consent, defining it as information generated by measurement of central or peripheral nervous system activity that a device can process. California followed the same year, treating neural data as sensitive personal information under the CCPA with opt-out rights, and explicitly excluding data merely *inferred* from non-neural sources such as heart rate.

Connecticut, Illinois, Massachusetts, Minnesota, Montana and Vermont have measures pending. No federal statute exists; senators requested FTC action in April 2025 without result.

Internationally, UNESCO adopted the [Recommendation on the Ethics of Neurotechnology](#) in November 2025, the first global standard in this area. It is non-binding, but it centres mental privacy, freedom of thought and autonomy, and specifically flags employment contexts in which consent is compromised by power asymmetry. Protocols should be designed against it, since review boards can be expected to cite it.

11.2 Design commitments adopted by this program

The state-decoding architecture argued for in §1.2 and §7.5 is itself the principal privacy safeguard: a system that reports dwell time and return latency cannot report what a person was thinking. Three further commitments follow.

Content decoding off by default. Where any content-sensitive component is used, it remains gated. Kunz et al. demonstrated a keyword gate in which decoding stays inactive until the user imagines a rare phrase, recognised at over 98% accuracy. The default state of the system is silence.

Cooperation as a structural protection. Tang & Huth demonstrated that decoding collapses to 0–3% under simple mental resistance. This is a property of the signal rather than a policy promise, and protocols should preserve rather than engineer around it.

Output belongs to the participant. The ARI is reported to the person it was measured from. The program does not produce third-party-facing inferences about individuals, and explicitly excludes employment, educational and insurance applications.

11.3 Invasive research: the appropriate pathway

Any implanted investigational device in a human requires an FDA Investigational Device Exemption. An intracortical array is a significant-risk device, so the IDE requires ISO 10993 biocompatibility data, electromagnetic compatibility testing, animal studies covering the intended implantation duration, a complete clinical protocol, documented manufacturing controls and IRB approval. Early feasibility studies typically enrol 3–10 participants. Self-experimentation constitutes no exception to any of these requirements, and data produced outside this framework is unpublishable.

The program therefore conducts no new implantation. Its intracranial phase relies on the established route: patients undergoing stereo-EEG monitoring for drug-resistant epilepsy, implanted for clinical reasons for one to two weeks, with research protocols added to consent. The relevant risk figure for the feasibility of any future invasive extension is from Section 8.4 — 3 of 14 BrainGate arrays never produced a consistent decoding signal.

11.4 Post-trial obligations

Participants in device trials have been left without access when trials conclude or sponsors withdraw, with no policy framework guaranteeing continued access and no obligation on manufacturers to maintain devices. Any protocol in this program that provides a participant with an ongoing training device specifies in advance what happens at study end, including continued access or a documented withdrawal plan agreed at enrolment.

12. Proposed research program

The program is structured so that each phase is independently publishable and earns the access the next phase requires. No new implantation occurs at any stage; the intracranial phase proceeds through existing clinical monitoring, and may prove unnecessary.

Three gates precede any intracranial work, and no new implantation occurs

Staged program, roughly three years, each phase funded by the last one working

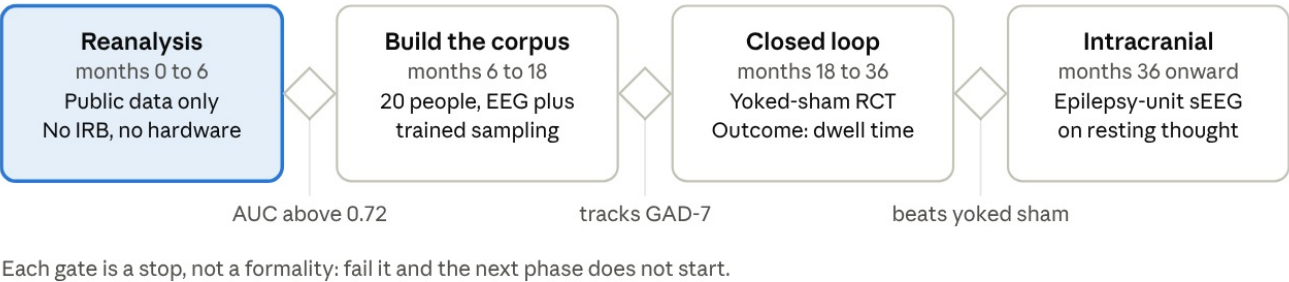


Figure 3. The staged program: four phases, three gates, no new implantation.

Figure 3. The staged program: four phases, three gates, no new implantation.

Phase 0 — Reanalysis (months 0–6)

Objective: establish that probe-reported mental state is decodable within individuals from scalp EEG, and that its decodability relates to anxiety.

ds001787 provides 24 participants, 64-channel EEG, with probes every two minutes rating concentration and mind-wandering. A within-subject classifier is trained on probe-reported state. The benchmark to exceed is $AUC \approx 0.72$ within individuals and ≈ 0.61 across them (Dong et al., 2021).

ds007216 then permits the question that has not been asked: 13 dimensions of thought content were rated at rest alongside STAI-S, GAD-7 and PHQ-9, so it is possible to test whether the decodability of self-referential thought covaries with trait anxiety. A positive result here is the empirical basis for the remainder of the program.

Phase 0 requires no IRB approval, since the data are public and de-identified, and no hardware.

Phase 1 — The Thought-Form Atlas (months 6–18)

Objective: construct the corpus that does not exist, and validate the ARI against it.

Twenty participants, high-density EEG, and Descriptive Experience Sampling with the multi-day iterative training the method requires. The target is sufficient paired neural and phenomenological data to train a classifier over the five sampled categories — inner speaking, inner seeing, unsymbolised thinking, feeling, sensory awareness — and to derive and calibrate the three ARI components per individual.

A deaf signing cohort is recruited in parallel, both to extend the Atlas to kinaesthetic inner signing and to establish baseline phenomenological distributions in a population for which, at present, no empirical data exist (\$7.3).

This phase also carries the autonomic sub-study testing temporal precedence (§10.3). It requires an IRB protocol but no device exemption. The Atlas is a citable resource independent of whether the closed-loop intervention succeeds.

Phase 2 — Closed-loop trial (months 18–36)

Objective: determine whether real-time feedback accelerates decentering beyond practice and expectancy.

Three arms, pre-registered: real feedback, yoked sham (another participant’s recording, matched for cue rate and reinforcement density), and practice-only. The primary endpoint is **dwelt time in decoded self-referential state**, not total time and not subjective calm, because dwell time is what the mechanism predicts and what self-report cannot inflate.

A four-week no-device follow-up is built into the original protocol, since a benefit that disappears with the hardware indicates substitution rather than skill acquisition.

This phase is where the program succeeds or fails. **Eight sham-controlled randomised trials of attention neurofeedback have found no significant pooled effect**, and a null result here, properly powered and pre-registered, would be a substantive contribution to a literature that has accumulated unblinded positive findings for two decades.

Phase 3 — Intracranial extension (months 36+)

Objective: characterise spontaneous inner experience intracranially, and test the signing-decoder prediction.

Conducted in partnership with an epilepsy monitoring unit running an active cognitive-neuroscience program. Patients implanted for clinical reasons for one to two weeks, cognitively intact, consented separately for research. The question is what spontaneous inner experience looks like intracranially when labelled by Descriptive Experience Sampling in a resting participant — and, in any signing participant, whether imagined signing decodes from hand-area contacts as §7.4 predicts.

This phase requires a named clinical collaborator from the outset. Institutions without a surgical program should initiate that partnership during Phase 1 rather than Phase 3, since access arrangements take years to establish and cost nothing to begin.

13. Hypotheses, endpoints and stopping rules

Stopping rules are specified in advance and pre-registered. A program with a strong prior commitment to its hypothesis will otherwise find reasons to continue past every ambiguous result.

#	HYPOTHESIS	PHASE	MEASURE	CONTINUE IF	STOP IF
H1	Probe-reported mind-wandering is decodable within-person from scalp EEG	0	Within-subject AUC on ds001787	AUC ≥ 0.75	AUC < 0.70 after fair tuning
H2	Decodability of self-referential thought covaries with trait anxiety	0	Per-subject dwell estimate vs GAD-7 on ds007216	r ≥ 0.30	95% CI excludes 0.30
H3	The five experience categories are separable within-person	1	Balanced accuracy, 5-class (chance 20%)	≥ 35%	≤ 25%
H4	Verbal rehearsal precedes and suppresses somatic arousal	1	Lagged cross-correlation, decoded verbal state vs SCR and HRV	Arousal falls 2–10 s after verbal onset	Arousal rises, or no lag structure
H5	Closed-loop feedback shortens dwell time vs yoked sham	2	Mean dwell duration in self-referential state	Between-group <i>d</i> ≥ 0.40 vs sham	CI includes 0
H6	The benefit survives removal of the device	2	Dwell time + Experiences Questionnaire at 4 weeks	Retained without hardware	Returns to baseline
H7	N1 suppression indexes how deliberate an inner utterance is	1	Auditory N1, matched vs mismatched probe, against rated deliberateness	Suppression scales with deliberateness	No relationship
H8	Rumination shows intact self-tagging but failed termination	1	N1 suppression (intact) alongside dwell time (long) in high-RNT individuals	Both hold together	Self-tag is degraded in ruminators too
H9	Imagined signing decodes from hand-area cortex better than imagined speech from orofacial cortex	3	Characters or signs per minute, accuracy	Matches or beats the 90 char/min handwriting baseline	Well below it
H10	Spontaneous experience is decodable intracranially at rest	3	5-class accuracy, DES-labelled, sEEG	Clear margin over scalp EEG	No margin

13.1 The decisive test

H5 is the program. Everything preceding it is instrumentation; everything following it is scale. If a closed-loop trainer cannot outperform a yoked sham on dwell time, the defensible conclusion is that the intervention adds nothing to unassisted practice.

H6 is the hypothesis most often omitted. A device that confers benefit only while worn has not taught metacognitive awareness; it has substituted for it. Decentering is expected to transfer, so the four-week no-device follow-up must be part of the original protocol rather than an addition.

13.2 Endpoints

- **Dwell time** in decoded self-referential state — primary endpoint, because the mechanism predicts it and self-report cannot inflate it
- **Return latency** — time from episode onset to attentional re-engagement; the second ARI component
- **Probe-caught fraction** — proportion of episodes the participant reports noticing *before* any external cue; the cleanest behavioural index that the skill is internalising, and expected to rise with training
- **Experiences Questionnaire** — the most widely used decentering measure, with the caveat that instruments in this family correlate only modestly with one another
- **GAD-7, PHQ-9, STAI-S** — already collected in ds007216, so Phase 0 and Phase 2 measures align

13.3 Falsification criteria for the program as a whole

Any one of the following should terminate the program rather than prompt a redesign:

1. Within-person state decoding plateaus near chance on high-quality scalp data across multiple independent datasets.
2. Decoded state shows no relationship to any anxiety or rumination measure in adequately powered samples.
3. Real feedback does not outperform yoked sham on dwell time in an adequately powered pre-registered trial.
4. Gains vanish within one month of device removal while a practice-only arm retains them.

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